

Boolean identities and laws 1

Advanced: Challenge 1



Boolean expressions can be manipulated using basic Boolean identities and laws. A Boolean identity is a statement of equivalence where the expression on the left of the equals sign has the same logic as the expression on the right.

Which of the following identities is **not** True?

- ☐ $A \vee 0 = A$
- ☐ $A \wedge \neg A = 0$
- ☐ $A \vee \neg A = 1$
- ☐ $A \wedge \neg A = 1$

Quiz:

STEM SMART 2026 Residential Extend Quiz 2 – Boolean Identities and Laws

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Boolean identities and laws 3

Advanced: Challenge 1



Boolean expressions can be manipulated using basic Boolean identities and laws. The following expressions are equivalent, which means that they implement that same Boolean logic:

$$A \wedge (B \wedge \neg C) = (A \wedge B) \wedge \neg C$$

What is the name of the law that has been applied?

- ☐ Absorption
- ☐ Commutativity
- ☐ Distribution
- ☐ Associativity

Quiz:

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Boolean identities and laws 5

Advanced: Challenge 1



Boolean expressions can be manipulated using basic Boolean identities and laws. The following expressions are equivalent, which means that they implement that same Boolean logic:

$$(A \wedge B) \vee (\neg A \wedge B) = B \wedge (A \vee \neg A)$$

What is the name of the law that has been applied?

- ☐ Distribution
- ☐ Commutativity
- ☐ Associativity
- ☐ Absorption

Quiz:

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De Morgan's Laws 1

Advanced: Challenge 1



Some Boolean expressions can be manipulated using De Morgan's laws. The expression below can be tackled in this way.

$$\neg(\neg A \vee \neg(B \wedge A))$$

Which of the following options shows a correct application of De Morgan's laws?

- ☐ $A \wedge B \vee A$
- ☐ $\neg A \vee \neg(B \wedge A)$
- ☐ $A \wedge B \wedge A$
- ☐ $A \wedge \neg B \wedge \neg A$

Quiz:

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De Morgan's Laws 2

Advanced: Challenge 1



Some Boolean expressions can be manipulated using De Morgan's laws. The expression below can be tackled in this way.

$$\neg(\neg B \wedge \neg(\neg B \wedge A))$$

Which of the following options shows a correct application of De Morgan's laws?

- ☐ $\neg B \vee A \vee B$
- ☐ $\neg B \vee A \wedge B$
- ☐ $\neg B \wedge A \vee B$
- ☐ $B \wedge A \vee B$

Quiz:

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De Morgan's Laws 3

Advanced: Challenge 2 

Human blood can be categorised based on the antigens and antibodies that it contains. There are four main blood types:

- Type O has no antigens
- Type A has A antigens
- Type B has B antigens
- Type AB has both A and B antigens

People with blood type O can receive donated blood with antigens that follow the rule below. Here variable A represents A antigens and variable B represents B antigens:

$$\neg A \wedge \neg B$$

Which of the following Boolean expressions is directly equivalent to the expression given above?

- ☐ $\neg(A \vee B)$
- ☐ $\neg(A \wedge B)$
- ☐ $\neg(\neg A \vee \neg B)$

Quiz:

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Simplify Boolean expression 4

Advanced: Challenge 1



Use the Boolean identities to simplify the Boolean expression below.

$$A \vee (B \wedge 0)$$

Which of the following options shows the simplest equivalent logic?

- ☐ A
- ☐ $A \vee B$
- ☐ $A \wedge B$
- ☐ 1
- ☐ B
- ☐ 0

Quiz:

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Simplify Boolean expression 3

Advanced: Challenge 1



Use the Boolean identities to simplify the Boolean expression below.

$$A \vee (B \wedge 1)$$

Which of the following options shows the simplest equivalent logic?

- ☐ A
- ☐ 0
- ☐ $A \vee B$
- ☐ $A \wedge B$
- ☐ B
- ☐ 1

Quiz:

STEM SMART 2026 Residential Extend Quiz 2 – Boolean Identities and Laws

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